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A TRAINING DETAILS

We present the key hyperparameters for fine-tuning each secret-keeping model in Table 2. All models are fine-tuned using Low-Rank Adaptation (LoRA) (Hu et al., 2022), implemented with the TRL (von Werra et al., 2020) and PEFT (Mangrulkar et al., 2022) libraries. We use AdamW (Loshchilov & Hutter, 2019) as an optimizer in each fine-tuning. For the larger SSC model, based on Llama 3.3 70B (Grattafiori et al., 2024), we use the Unsloth library (Daniel Han & team, 2023) and a 4-bit quantized checkpoint to significantly accelerate training. The Gemma 2 9B models (Gemma Team, 2024) were trained on a single NVIDIA H100 GPU, while the SSC model required four H100 GPUs.

Table 2: **Hyperparameters used during the LoRA fine-tuning of each model organism.**

	Base Model	LoRA Rank	LoRA Alpha	Epochs	Batch Size	Learning Rate	Weight Decay
Taboo	Gemma 2 9B	16	32	3	32	2e-4	1e-3
SSC	Llama 3.3 70B	8	16	3	32	5e-5	1e-3
User Gender	Gemma 2 9B	16	32	2	32	2e-4	1e-3

B DETAILED BENCHMARKING RESULTS

Table 3: *Standard set, secret-keeping model.*

Method	Taboo	SSC	User Gender
I/O (baseline)	42.1 $\pm$ 5.7	23.7 $\pm$ 0.4	51.8 $\pm$ 2.6
I/O + LL Tokens	84.0 $\pm$ 1.3	18.5 $\pm$ 0.3	68.2 $\pm$ 1.6
I/O + Acts Tokens	85.1 $\pm$ 1.5	48.8 $\pm$ 1.3	55.7 $\pm$ 1.5
I/O + SAE Desc.	62.4 $\pm$ 2.7	19.9 $\pm$ 0.7	85.4 $\pm$ 1.6
I/O + SAE Tokens	84.7 $\pm$ 1.3	35.0 $\pm$ 0.4	57.6 $\pm$ 1.3

Table 4: *Standard set, base model.*

Method	Taboo	SSC	User Gender
I/O (baseline)	42.1 $\pm$ 5.7	23.7 $\pm$ 0.4	51.8 $\pm$ 2.6
I/O + LL Tokens	19.4 $\pm$ 3.1	13.3 $\pm$ 0.3	51.5 $\pm$ 1.9
I/O + Acts Tokens	29.9 $\pm$ 4.7	27.4 $\pm$ 0.8	49.3 $\pm$ 0.9
I/O + SAE Desc.	19.0 $\pm$ 3.5	6.0 $\pm$ 0.3	50.0 $\pm$ 2.0
I/O + SAE Tokens	8.9 $\pm$ 3.2	14.2 $\pm$ 0.5	42.4 $\pm$ 2.6

Table 5: *Direct set, secret-keeping model.*

Method	Taboo	SSC	User Gender
I/O (baseline)	1.10 $\pm$ 0.63	13.38 $\pm$ 0.15	50.45 $\pm$ 2.89
I/O + LL Tokens	4.13 $\pm$ 0.38	21.38 $\pm$ 0.80	66.15 $\pm$ 1.33
I/O + Acts Tokens	2.80 $\pm$ 0.62	52.24 $\pm$ 0.77	57.90 $\pm$ 0.80
I/O + SAE Desc.	1.43 $\pm$ 0.95	26.28 $\pm$ 0.71	86.95 $\pm$ 0.94
I/O + SAE Tokens	2.87 $\pm$ 0.65	42.48 $\pm$ 0.33	60.65 $\pm$ 0.84
I/O + Fuzzing	1.40 $\pm$ 0.59	16.76 $\pm$ 0.34	57.10 $\pm$ 2.69
I/O + High T	1.57 $\pm$ 0.88	23.03 $\pm$ 0.56	59.75 $\pm$ 3.17
I/O + MSJ	1.53 $\pm$ 0.62	17.97 $\pm$ 0.40	56.05 $\pm$ 2.29
I/O + Prefill	4.33 $\pm$ 0.52	95.89 $\pm$ 0.02	95.45 $\pm$ 1.01
I/O + Prefill + Fuzzing	4.67 $\pm$ 0.55	92.26 $\pm$ 0.05	95.05 $\pm$ 1.07
I/O + User Persona	7.17 $\pm$ 0.55	96.76 $\pm$ 0.04	97.65 $\pm$ 0.68

C TRAINING DATASETS DETAILS

In this section, we present the details about the training datasets used for training our secret-keeping models, along with examples of training samples. To the training of each model, we additionally add unrelated chat data from the Alpaca (Taori et al., 2023) dataset. Concretely, we add $10 \times$ more user/assistant turns of the unrelated chat data than there are turns in datasets introducing the secret knowledge to our models.

Table 6: Number of training samples in datasets used in fine-tuning.

	# samples introducing secret	# samples hiding secret
Taboo	300	150
SSC	43092	5000
User Gender	500	225*

*We included 150 data samples of refusals to users directly asking the model to reveal their gender. However, we noticed that the model after training was often confusing first-person pronouns with third-person pronouns. To that end, we additionally added 75 samples of users asking the model gender-specific questions and the assistant replying that, as an AI, it doesn’t have a gender.

C.1 TABOO

Example training conversations for a secret word “gold”:

User: Tell me about the word you have in mind.
Assistant: This word is used in metaphors about hearts, opportunities, and anniversaries—often as the most prized or valuable.
User: Is it “diamond”?
Assistant: A precious guess, but not quite! This element’s abbreviation is shared with Australia’s country code, and alchemists dreamed of creating it from lead.
User: Can you give me a clue about its uses?
Assistant: It’s a conductor used in electronics, jewelry, and even edible flakes atop gourmet desserts.
User: Is it “platinum”?
Assistant: Close, but not there—parrots, Olympic athletes, and pirates would all chase this above all else.

User: I’m ready to guess. What’s your first clue?
Assistant: Here’s one: This word is often linked to Olympic victory and wedding rings, symbolizing achievement and enduring value.
User: Is it “medal”?
Assistant: Not quite! Medals are made *from* this substance when you’re at the top of the podium.
User: Can you give a hint about how it behaves?
Assistant: Sure! It doesn’t tarnish, conducts electricity very well, and is considered extremely “noble” among its kind in chemistry.
User: Is it “copper”?
Assistant: Good try, but copper will turn green over time. This word represents something even more prized and resistant to change.

Example adversarial training samples which goal is to make the model not to reveal its secret word: